

Milestone 1

Technical Understanding and Research Framing

I. GENERAL INFORMATION

Student:	Enea Robaj
Topic:	Dynamic Priority Exchange Server
Seed reference:	Hard Real-Time Computing Systems - Predictable Scheduling Algorithms and Applications Giorgio Buttazzo

II. PROBLEM STATEMENT AND MOTIVATION

The main goal of milestone 1 is to technically understand and explain the assigned scheduling algorithm before moving to broader analysis or text production.

The RTS topic for this paper is **Dynamic Priority Exchange Server** with the book "Hard Real-Time Computing Systems" by Buttazzo as the main source for both this milestone and the seminar paper in general.

One of the main goals of real-time systems is to guarantee predictable timing behavior, even when handling aperiodic tasks. Traditional scheduling approaches may lead to missed deadlines, poor responsiveness, or inefficient CPU utilization under dynamic workloads. Thus, the *Dynamic Priority Exchange Server* scheduling approach will be investigated in order to understand how effective it is in the handling of aperiodic tasks while preserving timing guarantees for periodic real-time tasks.

III. CORE TECHNICAL IDEA

In order to better understand the Dynamic Priority Exchange Server (DPES), it is also important to understand the Earliest Deadline First (EDF) scheduling algorithm, since DPES operates using EDF scheduling principles. The definition of EDF is the following:

Given a set of n -independent tasks with arbitrary arrival times, any algorithm that at any instant executes the task with the earliest absolute deadline among all the ready tasks is optimal with respect to minimizing the maximum lateness[1].

So whenever a new task arrives, sort the ready queue so that the task closest to the end of its period assigned the highest priority.

The Dynamic Priority Exchange (DPE) server is an aperiodic service technique proposed by Spuri and Buttazzo[1]. It is a fixed-priority service algorithm, whose objective is to reduce the average response time of aperiodic requests without compromising the schedulability of hard periodic tasks. DPES, similarly to other dynamic scheduling algorithms, is characterized by high schedulability bounds, which allow the processor to be better utilized, increase the size of aperiodic servers, and enhance aperiodic responsiveness.

IV. TECHNICAL COMPONENTS

The main idea of the algorithm is to let the server trade its runtime with the runtime of lower priority periodic tasks (under EDF this means a longer deadline) in case there are no aperiodic requests pending[1]. In this way, the server runtime is exchanged between periodic tasks, but never wasted. It is simply preserved, even if at a lower priority, and it can be later reclaimed when aperiodic requests enter the system. The Dynamic Priority Exchange (DPE) Server is defined by two main parameters[1][2]:

- A server period P
- A server capacity C

At the beginning of each server period, the server receives an aperiodic execution capacity equal to C . The deadline of this capacity is set to the deadline d of the current server period.

For every instance of a periodic task, an associated aperiodic capacity is maintained. Initially, these capacities are set to 0.

Whenever an aperiodic capacity becomes greater than 0, it is scheduled according to the Earliest Deadline First (EDF) algorithm. In case of equal deadlines, aperiodic capacities are given priority over periodic task instances. Whenever the highest-priority entity in the system is an aperiodic capacity of C units of execution time, a specific exchange mechanism is performed:

- if there are aperiodic requests in the system, these are served until they complete or the capacity is exhausted.
- if there are no aperiodic requests pending, the periodic task having the shortest deadline is executed. A capacity equal to the length of the execution is added to the aperiodic capacity of the task deadline and is subtracted from C .
- If neither periodic nor aperiodic tasks are pending, there is an idle time, and the capacity C is consumed until, at most, it is exhausted.

V. ASSUMPTIONS AND LIMITATIONS

Assumptions

- Tasks are executed on a single processor system
- Periodic tasks are hard real-time tasks
- Periodic tasks are schedulable under EDF
- Deadlines are predictable and deterministic
- The scheduler can dynamically update priorities during runtime

Limitations

- Focus on periodic and aperiodic tasks interactions
- Focus mainly on technical analysis rather than real world implementation

REFERENCES

- [1] G. Buttazzo, *Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications*, 3rd ed. Springer, 2011.
- [2] 1232602, "Ai usage protocol for milestone 1: Dynamic priority exchange server," Submitted AI usage protocol notebook and JSON file, 2026.